

Ritabrata Ghosh

📍 Bangalore, India ✉ ritabrataghosh09@gmail.com ☎ +91 7365956148 📁 Portfolio in Ritabrata Ghosh
🔗 RitoG09

Skills

Languages: C/C++, JavaScript, TypeScript

Technologies: Node.js, Next.js, React.js, Express.js, PostgreSQL, MongoDB, Prisma, AWS, Docker, CI/CD

Coursework: Data Structures Algorithms (DSA), Object Oriented Programming, Discrete Maths, Operating Systems (OS), Database Management Systems (DBMS), Computer Networks

Education

JSSATE Bangalore

B.E. in Computer Science and Engineering

Aug 2023 - June 2027

CGPA:8.5/10

Experience

Open Source Contributor, JSON Schema Organization

Nov 2025 - Present

- Contributed to **json-schema-org**, **Sourcemeta**, and **Hyperjump** by improving JSON Schema validation workflows, CLI tooling, and developer experience.
- Enhanced **error categorization**, **lint ordering by source location**, and **exit code classification** to ensure standards-compliant schema validation behavior.
- Implemented features such as **pagination**, **state persistence**, search state management across actively maintained production-scale repositories.

Projects

CollabDraw: Real-Time Collaborative Whiteboard



(Next.js, Node.js, WebSockets, Kafka, PostgreSQL, Turborepo, Rough.js)

- Developed a real-time collaborative whiteboard using **WebSockets** with scalable room-based architecture and synchronized state management.
- Integrated **Kafka** for event-driven, low-latency chat messaging within sessions for scalability.
- Implemented drawing engine with **Rough.js** and structured a scalable monorepo using **Turborepo**.

BrainDoc: AI-Powered Document Intelligence Platform



(Next.js, Node.js, PostgreSQL, PineconeDB, RAG Pipeline, OpenAI API, YouTube Data API)

- Built an AI-driven document query system using a **Retrieval-Augmented Generation (RAG)** pipeline for context-aware responses.
- Leveraged **Pinecone vector database** for semantic search and scalable embedding retrieval.
- Designed full-stack architecture with persistent chat history and contextual video retrieval integration.

Planova: AI-Powered Travel Planning System



(React, Node.js, Express, MongoDB, Gemini API, Google Places API, REST APIs)

- Developed an AI-driven itinerary generation system using **Gemini API** for personalized, constraint-aware travel planning.
- Integrated **Google Places API** for real-time location insights and hotel recommendations.

Achievements

- 1st Place: SJBIT Hackathon AXIOM 2025** - Won 50,000 prize for developing an innovative hardware based healthcare solution.
- 1st Place: Dr. TTIT Nexathon 2025** - Secured 20,000 prize for building a high-impact technical prototype under time constraints.
- 2nd Place: Global Ideathon 2025** - Awarded 15,000 prize for presenting a scalable and practical innovation concept.
- 3rd Place: Google Developer Groups Ideathon**